LAB - 6

Huffman Coding & Job Sequencing Problem

Huffman Coding:

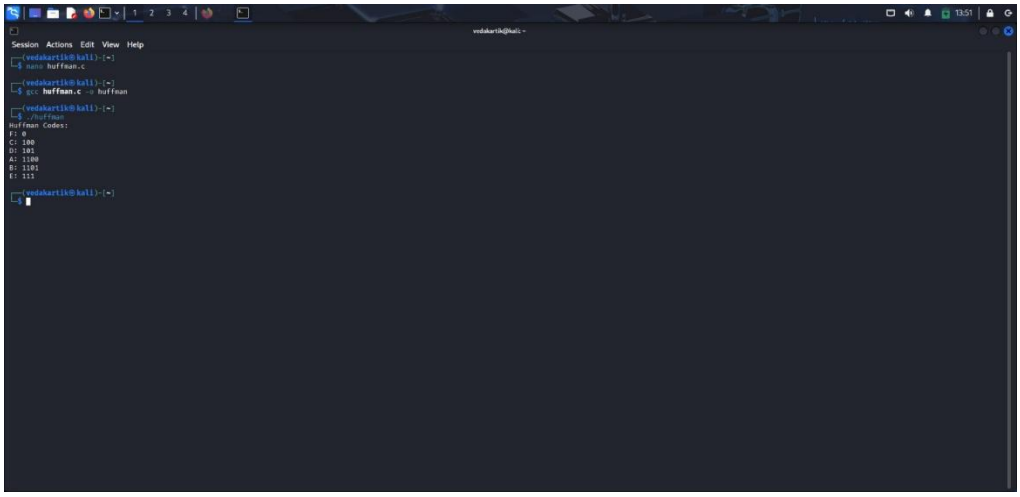
CODE:

```
#include <stdio.h>
#include <stdlib.h>
#define MAX 100

struct Node{
char data;
unsigned freq;
   struct Node *left,*right;
};

struct Node* newNode(char data,unsigned freq){     struct Node*
temp=(struct Node*)malloc(sizeof(struct Node));     temp-
>left=temp->right=NULL;     temp->data=data;     temp-
>freq=freq;     return temp;
}

struct MinHeap{
unsigned size;     struct
Node* arr[MAX];
};

void swap(struct Node** a,struct Node** b){
struct Node* t=*a;
   *a=*b;
   *b=t;
}

void heapify(struct MinHeap* heap,int i){
int smallest=i;     int left=2*i+1;     int
right=2*i+2;

   if(left<heap->size && heap->arr[left]->freq<heap->arr[smallest]->freq)
smallest=left;

   if(right<heap->size && heap->arr[right]->freq<heap->arr[smallest]->freq)
smallest=right;
```

```
    if(smallest!=i){        swap(&heap->arr[i],&heap-
>arr[smallest]);        heapify(heap,smallest);
    }
}

struct Node* extractMin(struct MinHeap* heap){
struct Node* temp=heap->arr[0];     heap-
>arr[0]=heap->arr[--heap->size];
heapify(heap,0);     return temp;
}

void insertHeap(struct MinHeap* heap,struct Node* node){
int i=heap->size++;     while(i && node->freq<heap->arr[(i-
1)/2]->freq){        heap->arr[i]=heap->arr[(i-1)/2];        i=(i-
1)/2;
    }
    heap->arr[i]=node;
}

struct Node* buildTree(char data[],int freq[],int size){
struct MinHeap heap;     heap.size=0;

    for(int i=0;i<size;i++)
heap.arr[heap.size++]=newNode(data[i],freq[i]);

    for(int i=(heap.size-1)/2;i>=0;i--)
heapify(&heap,i);

    while(heap.size>1){        struct Node*
left=extractMin(&heap);        struct Node*
right=extractMin(&heap);

        struct Node* top=newNode('$',left->freq+right->freq);
top->left=left;        top->right=right;

        insertHeap(&heap,top);
    }
    return extractMin(&heap);
}

void printCodes(struct Node* root,int arr[],int top){
if(root->left){        arr[top]=0;
printCodes(root->left,arr,top+1);
    }
```

```
    if(root->right){         arr[top]=1;
printCodes(root->right,arr,top+1);
    }
    if(!root->left && !root->right){
printf("%c: ",root->data);
for(int i=0;i<top;i++)
printf("%d",arr[i]);         printf("\n");
    }
}

int main(){     char
data[]={'A','B','C','D','E','F'};     int
freq[]={5,9,12,13,16,45};     int
size=sizeof(data)/sizeof(data[0]);

    struct Node* root=buildTree(data,freq,size);

    int             arr[MAX],top=0;
printf("Huffman     Codes:\n");
printCodes(root,arr,top);

    return 0;
}
```

OUTPUT:



Job Sequencing Problem

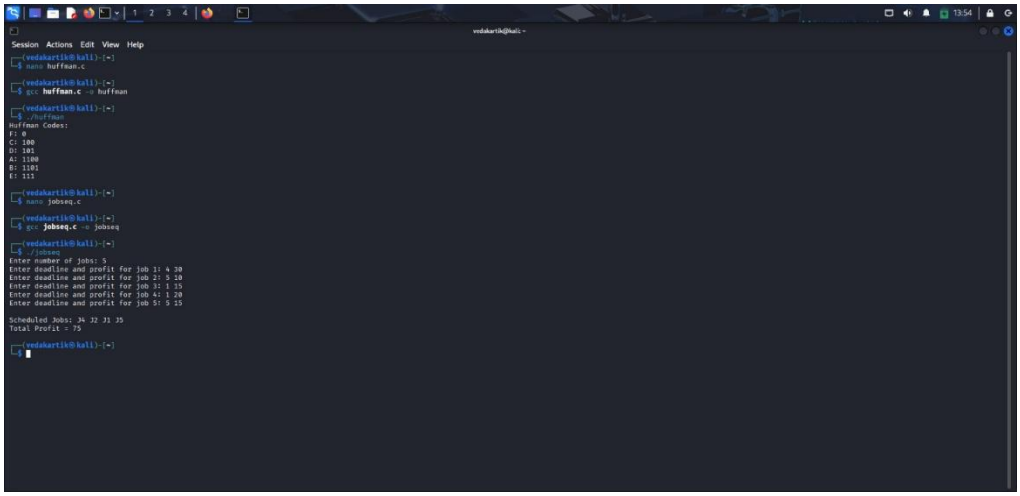
CODE:

```
 #include <stdio.h>
```

```
#include <stdlib.h>

typedef struct {
int id; int
deadline; int
profit;
} Job;

int compare(const void *a, const void *b) {
Job *j1 = (Job *)a; Job *j2 =
(Job *)b; return j2->profit -
j1->profit;
}

int main() { int n; printf("Enter
number of jobs: "); scanf("%d",
&n);

Job jobs[n];

for(int i = 0; i < n; i++) { printf("Enter deadline and
profit for job %d: ", i+1); scanf("%d %d",
&jobs[i].deadline, &jobs[i].profit); jobs[i].id = i + 1;
}

qsort(jobs, n, sizeof(Job), compare);

int maxDeadline = 0;
for(int i = 0; i < n; i++)
if(jobs[i].deadline > maxDeadline)
maxDeadline = jobs[i].deadline;

int slot[maxDeadline]; for(int i =
0; i < maxDeadline; i++) slot[i] =
-1;

int totalProfit = 0;

for(int i = 0; i < n; i++) { for(int j =
jobs[i].deadline - 1; j >= 0; j--) { if(slot[j]
== -1) { slot[j] = jobs[i].id; totalProfit +=
jobs[i].profit; break;
}
}
```

```
}


printf("\nScheduled Jobs: ");
for(int i = 0; i < maxDeadline;
i++) if(slot[i] != -1) printf("J%d ",
slot[i]);

printf("\nTotal Profit = %d\n", totalProfit);

return 0;
}
```

OUTPUT:



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